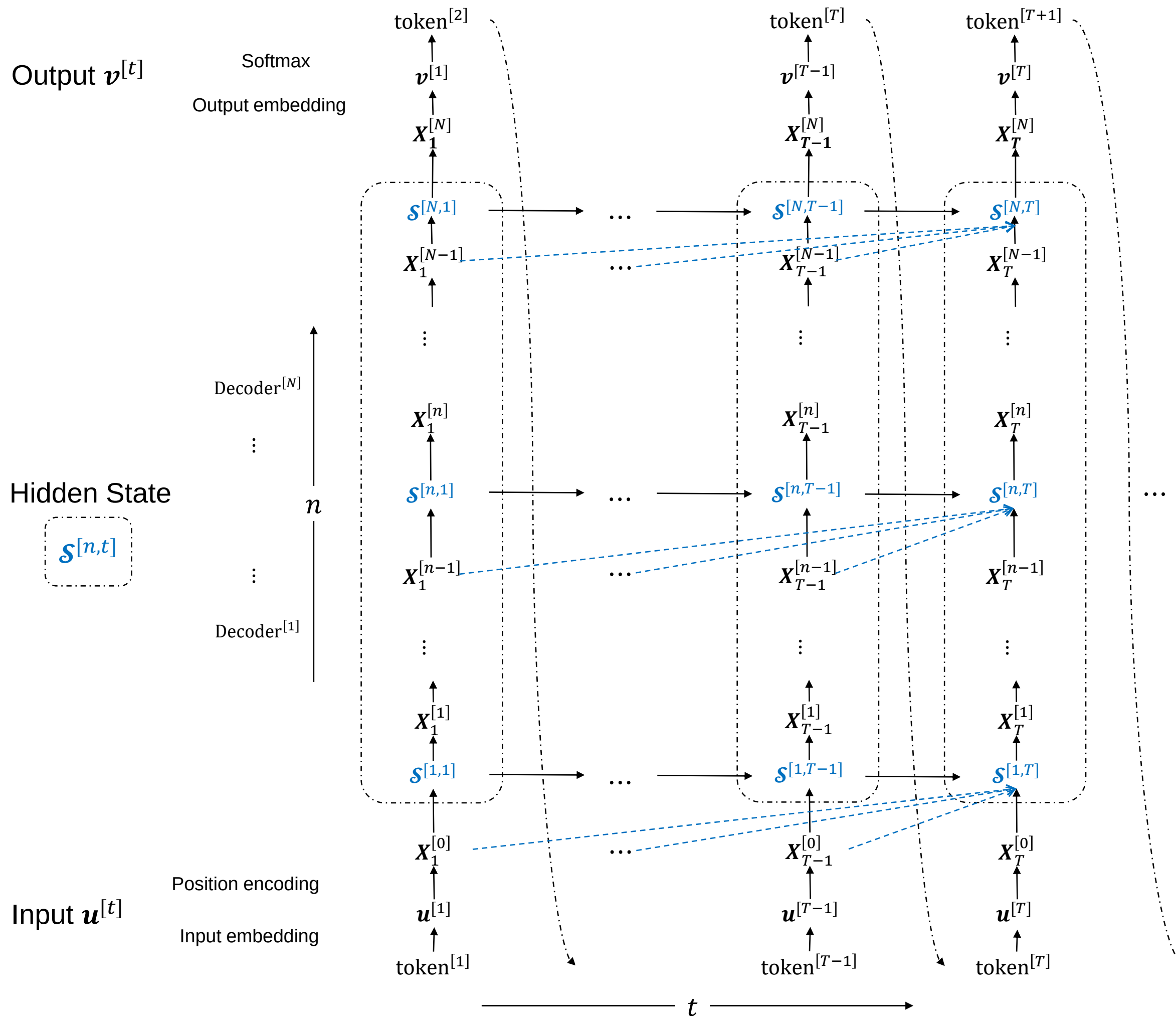




Update rule

$$\begin{aligned}\mathcal{S}^{[n,T]} &= F\left(\mathcal{S}^{[n,T-1]}, X_T^{[n-1]}\right) = \mathcal{S}^{[n,T-1]} \circ X_T^{[n-1]} \\ &= \mathcal{S}^{[n,0]} \circ X_1^{[n-1]} \circ X_2^{[n-1]} \circ \dots \circ X_T^{[n-1]}\end{aligned}$$

◦ 运算满足结合律即可序列并行

$$X_T^{[n]} = O(\mathcal{S}^{[n,T]}, X_T^{[n-1]})$$

Full Att.

$$\mathcal{S}^{[n,T]} = \mathcal{S}^{[n,T-1]}.append\left(X_T^{[n-1]}\right)$$

Linear Att./SSM/...

$$\mathcal{S}^{[n,T]} = \mathcal{S}^{[n,T-1]} \odot M\left(X_T^{[n-1]}\right) + VK(X_T^{[n-1]})$$